Notes on Varieties of Semiring Solvers

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October 11, 2026

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1 Category theory basics

Definition 1 (Category)

A *category* $\mathcal{C}$ consists of:

- A class of objects $\mathrm{Ob}(\mathcal{C})$,
- For each pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, a class of morphisms $\mathrm{Hom}_{\mathcal{C}}(a, b)$,
- For each object $a \in \mathrm{Ob}(\mathcal{C})$, an identity morphism $\mathrm{id}_a \in \mathrm{Hom}_{\mathcal{C}}(a, a)$,
- A composition operation $\circ$ such that for morphisms $f : a \to b$ and $g : b \to c$, there is a morphism $g \circ f : a \to c$,

satisfying the following axioms:

- Associativity: for morphisms $f : a \to b$, $g : b \to c$, and $h : c \to d$, we have $h \circ (g \circ f) = (h \circ g) \circ f$.
- Identity: for any morphism $f : a \to b$, we have $\mathrm{id}_b \circ f = f = f \circ \mathrm{id}_a$.

Definition 2 (Subcategory)

A *subcategory* $\mathcal{D}$ of a category $\mathcal{C}$ consists of:

- A class of objects $\mathrm{Ob}(\mathcal{D}) \subseteq \mathrm{Ob}(\mathcal{C})$,
- For each pair of objects $a, b \in \mathrm{Ob}(\mathcal{D})$, a class of morphisms $\mathrm{Hom}_{\mathcal{D}}(a, b) \subseteq \mathrm{Hom}_{\mathcal{C}}(a, b)$,

such that the identity morphisms and composition in $\mathcal{D}$ are inherited from $\mathcal{C}$.

Definition 3 (Subcategory classes)

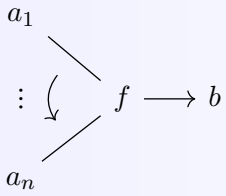
A subcategory $\mathcal{D}$ of a category $\mathcal{C}$ is called:

- *full* if for all objects $a, b \in \mathrm{Ob}(\mathcal{D})$, we have $\mathrm{Hom}_{\mathcal{D}}(a, b) = \mathrm{Hom}_{\mathcal{C}}(a, b)$,
- *wide* if $\mathrm{Ob}(\mathcal{D}) = \mathrm{Ob}(\mathcal{C})$.
- *replete* if for every object $a \in \mathrm{Ob}(\mathcal{C})$ that is isomorphic to an object in $\mathrm{Ob}(\mathcal{D})$, we have $a \in \mathrm{Ob}(\mathcal{D})$.

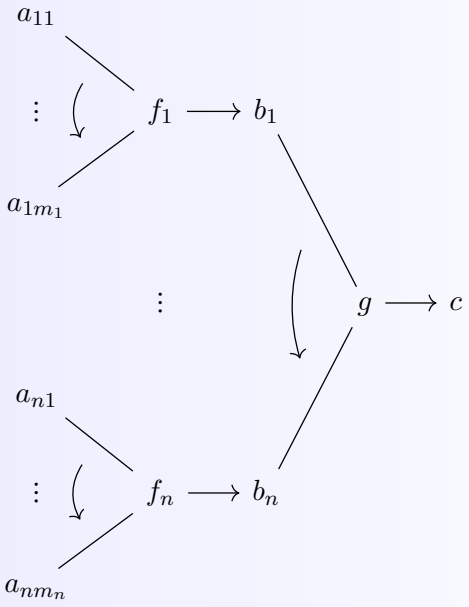
Definition 4 (Multicategory)

A *multicategory* $\mathcal{U}$ consists of:

- A collection $\mathrm{Ob}(\mathcal{U})$ of *objects*.
- A collection $\mathbb{U}(a_1, \ldots, a_n; b)$ of *multi-morphisms*, for every (possibly empty) list of *source* objects $a_1, \ldots, a_n$ and *target* object b, in which case we write $f : (a_1, \ldots, a_n) \to b$ and depict the multi-morphisms like so:



  The *morphisms* are those multi-morphisms whose source is of length 1.
- An *identity* morphism $\mathrm{id}_a : (a) \to a$ for each object a.
- The *composition* operation, defined for every multi-morphism $g : (b_1, \ldots, b_n) \to c$ and compatible list of multi-morphisms $f_i : (a_{i1}, \ldots, a_{im_i}) \to b_i$, for each $1 \leqslant i \leqslant n$, yielding a multi-morphism $g \circ (f_1, \ldots, f_n) : (a_{11}, \ldots, a_{1m_1}, \ldots, a_{n1}, \ldots, a_{nm_n}) \to c$. We depict the composed morphism by pasting the shared objects:



  such that:

  – identities are neutral, i.e., for all $f : (a_1, \ldots, a_n) \to b$:

$$f \circ (\mathrm{id}_{a_1}, \ldots, \mathrm{id}_{a_n}) = f = \mathrm{id}_b \circ f$$

  – composition is associative:

$$h \circ (g_1 \circ (f_{11}, \ldots, f_{1m_1}), \ldots, g_n \circ (f_{n1}, \ldots, f_{nm_n})) = (h \circ (g_1, \ldots, g_n)) \circ (f_{11}, \ldots, f_{1m_1}, \ldots, fn1, \ldots, f_{nm_n})$$

diagram of the associativity

Definition 5 (Operad)

An *operad* $\mathcal{O}$ consists of:

- A collection of sets $\{\mathcal{O}_n\}_{n\geq 0}$, where $\mathcal{O}_n$ is the set of n-ary operations,
- A composition function $\gamma : \mathcal{O}_n \times \mathcal{O}_{k_1} \times \cdots \times \mathcal{O}_{k_n} \to \mathcal{O}_{k_1+\cdots+k_n}$,
- An identity element $e \in \mathcal{O}_1$,

satisfying the following axioms:

- Associativity: for all $f \in \mathcal{O}_n$, $g_i \in \mathcal{O}_{k_i}$, and $h_{ij} \in \mathcal{O}_{m_{ij}}$, we have

$$\gamma(f; g_1, \ldots, g_n) = f(g_1(h_{11}, \ldots, h_{1k_1}), \ldots, g_n(h_{n1}, \ldots, h_{nk_n})).$$

- Identity: for all $f \in \mathcal{O}_n$, we have

$$f(e, \ldots, e) = f.$$

2 Verbal categories

Example 6

The category $\mathbb{F}$ of finite ordinals consists of:

- Objects: the finite ordinals $[n] = \{0, \ldots, n-1\}$ for $n \in \mathbb{N}$.
- Morphisms: the functions between finite ordinals.

Let E, W and C respectively denote the structural rules of exchange, weakening and contraction. Allowing the rules designated below on the left corresponds to the subcategories of $\mathbb{F}$ on the right:

$$\begin{array}{ccccc} \varnothing & \longrightarrow & \{\mathrm{E}\} & \longrightarrow & \{\mathrm{E},\mathrm{C}\} \\ \downarrow & & \downarrow & & \downarrow \\ \{\mathrm{W}\} & \longrightarrow & \{\mathrm{E},\mathrm{W}\} & \longrightarrow & \{\mathrm{E},\mathrm{W},\mathrm{C}\} \end{array} \qquad \begin{array}{ccccc} \mathbb{F}_{\mathrm{id}} & \longrightarrow & \mathbb{F}_{\mathrm{bij}} & \longrightarrow & \mathbb{F}_{\mathrm{surj}} \\ \downarrow & & \downarrow & & \downarrow \\ \mathbb{F}_{m,\mathrm{inj}} & \longrightarrow & \mathbb{F}_{\mathrm{inj}} & \longrightarrow & \mathbb{F} \end{array}$$

Definition 7 (Chosen coproducts in $\mathbb{F}$)

Let $n \in \mathbb{N}$, $\vec{m} = (m_i \in \mathbb{N})_{i\in n}$ and $m = \sum_{i\in n} m_i$. For each $i \in n$, we define the function

$$\iota_{\vec{m},i} : m_i \to m$$

by

$$\iota_{\vec{m},i}(j) = \left(\sum_{k\in i} m_k\right) + j$$

for each $j \in m_i$. The family $(\iota_{vecm,i} : m_i \to m)_{i\in n}$ exhibits m as a coproduct of $\vec{m}$ in $\mathbb{F}$.

Definition 8 (Verbal substitution operation)

Given morphisms $(\beta : n' \to n, \vec{\alpha} = \alpha_i : m'_i \to m_i)_{i \in n}$ in $\mathbb{F}$, define $m' = \sum_{i' \in n'} m'_{\beta(i')}$ and $m = \sum_{i \in n} m_i$. We define

$$\beta \star \vec{\alpha} : m' \to m$$

as the unique morphism in $\mathbb{F}$ making the following square commute for each $i' \in n'$.

$$\begin{array}{ccc} m'_{\beta(i')} & \xrightarrow{\alpha_{\beta(i')}} & m_{\beta(i')} \\ {\scriptstyle \iota_{\vec{m}'\beta,i'}}\downarrow & & \downarrow{\scriptstyle \iota_{\vec{m},\beta(i')}} \\ m' & \xrightarrow[\beta\star\vec{\alpha}]{} & m \end{array}$$

Here, $\vec{m}'\beta = (m'_{\beta(i')})_{i' \in n'}$ and $\vec{m} = (m_i)_{i \in n}$.

Definition 9 (Verbal category, [Tro02])

A *verbal category* $\mathbb{W}$ is a wide subcategory of $\mathbb{F}$ that is closed under the verbal substitution operation $\beta \star \vec{\alpha}$ of Def 8.
We say that a verbal category is *symmetric* if it contains $\mathbb{F}_{\text{bij}}$.

In the rest of this section, $\mathbb{W}$ denotes a verbal category.

Definition 10 ($\mathbb{W}$-multicategory, [Tro02])

References

[Tro02] S.N. Tronin. "Abstract clones and operads." eng. In: *Sibirskij Matematicheskij Zhurnal* 43.4 (2002), 924–936 (2002), translation in Sib. Math. J. 43. URL: http://eudml.org/doc/50186.